R29
10k

MCU Connections

U1
STM32C542CCT6

alert pins from the power monitoring

GATE10 1 PE2
2 PC13 Low current/freq
3 PC14-OSC32_IN Low current/freq
4 PC15-OSC32_OUT Low current/freq
OSC_IN 5 PH0-OSC_IN
OSC_OUT 6 PH1-OSC_OUT
nRST 7 NRST
8 VREF- Not using DAC/ADC
9 VREF+ GND VREF+?
GATE1 10 PA0
GATE2 11 PA1
GATE3 12 PA2
GATE4 13 PA3
PI_MOSI 14 PA4
SCLK 15 PA5
PI_MISO 16 PA6
GATE5 17 PA7
GATE6 18 PB0
GATE7 19 PB1
GATE8 20 PB2
GATE9 21 PB10
22 VCAP
23 VSS
24 VDD

48 VDD
47 VSS
46 PB9 STM_TO_INPUT_CAN
45 PB8 STM_TO_OUTPUT_CAN
44 PH2-BOOT0
43 PB7 PWM_B2
42 PB6 PWM_C1
41 PB5 PWM_C2
40 PB4 PWM_D1
39 PB3 PWM_D2
38 PA15 CS_BN0
37 PA14 SWCLK
36 VDD
35 VSS
34 PA13 SWDIO
33 PA12 SDA
32 PA11 SCL
31 PA10 CS_BMI
30 PA9 DEBUG
29 PA8 PWM_A1
28 PB15 PWM_B1
27 PB14 nINT_BN0 GPIO
26 PB13 RST_BN0 GPIO
25 PB12 INT1_BMI GPIO

PWM (Bitbang GPIO)

3.3V
C1 0.1u
R1 10k
R2 10k
C2 0.1u
C3 0.1u
C20 2.2u
C4 0.1u
C5 4.7u
ADC? (Prob no)

Categorized Designs

MOSFET_driver
File: untitled.kicad_sch

Sensors_Telemetry
File: Sensors_Telemetry.kicad_sch

Connector_Signals
File: connector_signals.kicad_sch

Voltage Conv-Protec-Monitoring
File: Voltage_conv_prot.kicad_sch

Crystal Oscillator

Y2
ECS-240-8-37B-CKY-TR
R3
0
OSC_IN
OSC_OUT
C6 8p
C7 8p
NEW CAP VALUES

Test Points, LEDs

TP2 GND_TP
TP1 12V_TP
TP3 5.2V_TP
TP4 3.3VTP
DEBUG
12V
5.2V
3.3V
D15 RED
D1 BLUE
D2 GREEN
D3 RED
R34 220
R4 440
R5 310
R6 65
+12V
+5.2V
+3.3V

Add LED debug

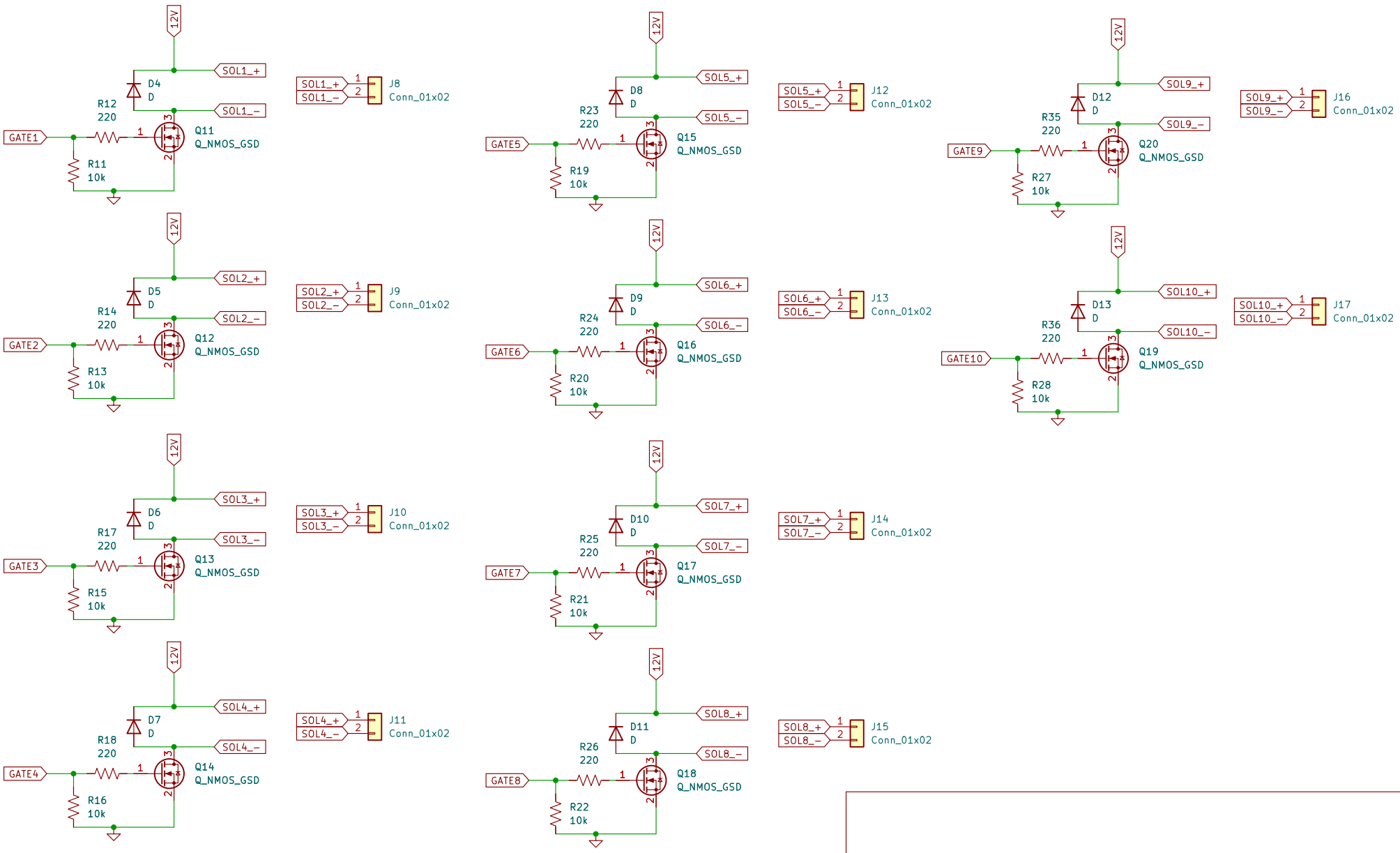
20mA for Blue (3.3V), Red (2.0V)
10mA for Green (2.1V)

CAN Transceiver

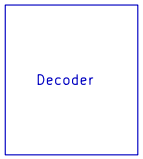
U2
TCAN1044VDRQ1
CR1
ESD2CAN24DBZRQ1

5.2V
3.3V
C19 0.1u
C8 0.1u
3 VCC
5 VIO
STM_TO_INPUT_CAN 1 TXD
8 STB
STM_TO_OUTPUT_CAN 4 RXD
7 CANH
6 CANL
2 GND
R7 60
R8 60
C9 0.1u
CAN_H
CAN_L

Purdue ROV		
Sheet: /		
File: X19-Control-Board.kicad_sch		
Title: Purdue ROV Subsystem Board		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5-unknown		Id: 1/6


12V
SOL1_+
SOL1_−
D4
D
R12
220
GATE1
Q11
Q_NMOS_GSD
R11
10k
J8
Conn_01x02
SOL2_+
SOL2_−
D5
R14
GATE2
Q12
R13
J9
SOL3_+
SOL3_−
D6
R17
GATE3
Q13
R15
J10
SOL4_+
SOL4_−
D7
R18
GATE4
Q14
R16
J11
SOL5_+
SOL5_−
D8
R23
GATE5
Q15
R19
J12
SOL6_+
SOL6_−
D9
R24
GATE6
Q16
R20
J13
SOL7_+
SOL7_−
D10
R25
GATE7
Q17
R21
J14
SOL8_+
SOL8_−
D11
R26
GATE8
Q18
R22
J15
SOL9_+
SOL9_−
D12
R35
GATE9
Q20
R27
J16
SOL10_+
SOL10_−
D13
R36
GATE10
Q19
R28
J17
Sheet: /MOSFET_driver/
File: untitled.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5−unknown
Id: 2/6



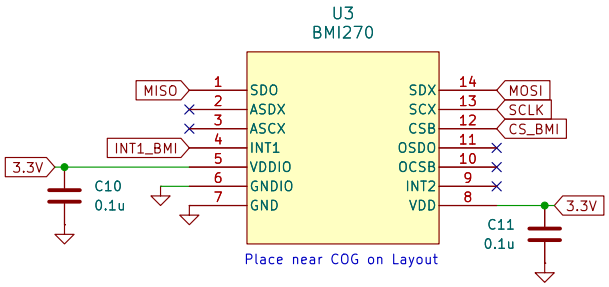


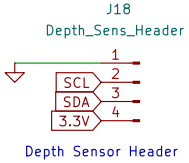


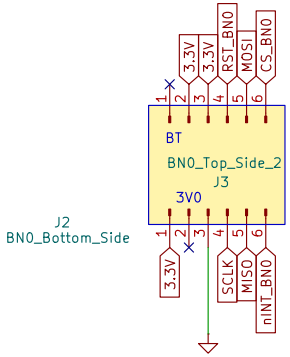
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File: temp_idea_555.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5-unknown		Id: 3/6



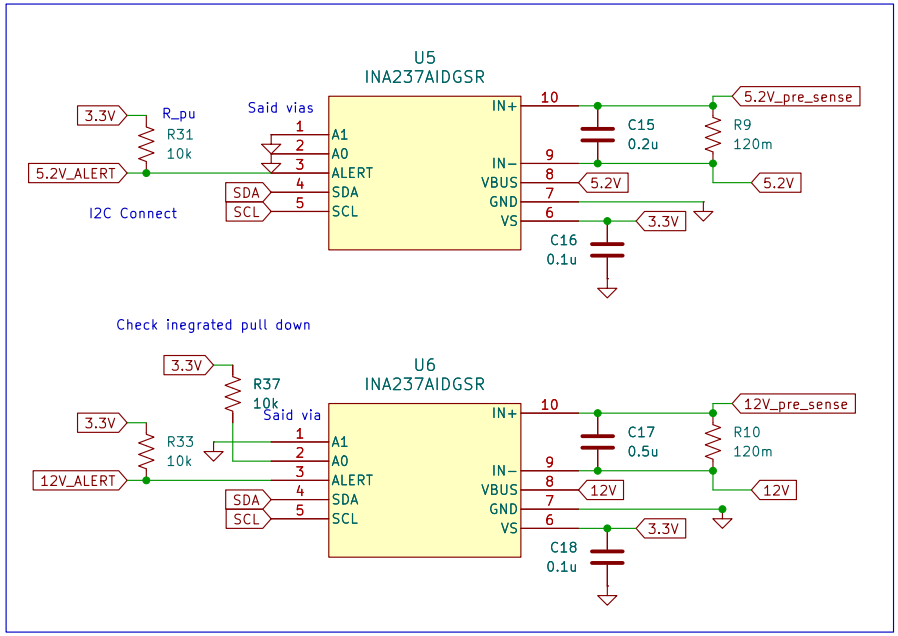




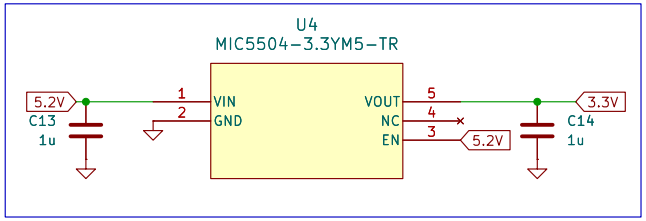
Making a symbol for this next...

Sheet: /Sensors_Telemetry/		
File: Sensors_Telemetry.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5-unknown		Id: 5/6

Power Monitoring 5.2V, 12V rails



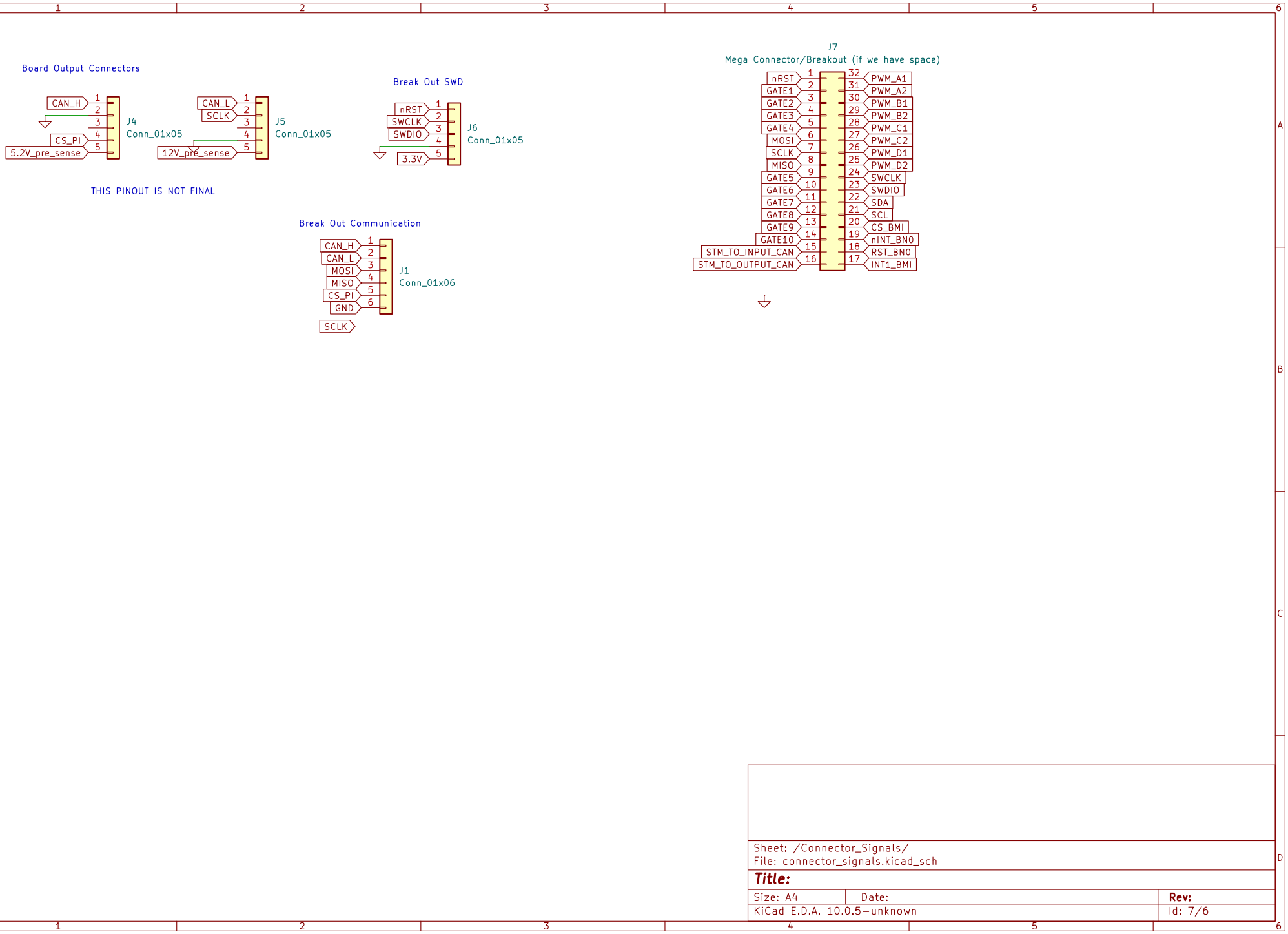
LDO – Voltage Conversion



Sheet: /Voltage Conv-Protec-Monitoring/
File: Voltage_conv_prot.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5-unknown		Id: 6/6


Board Output Connectors
CAN_H
CS_PI
5.2V_pre_sense
J4
Conn_01x05
CAN_L
SCLK
12V_pre_sense
J5
Conn_01x05
THIS PINOUT IS NOT FINAL
Break Out SWD
nRST
SWCLK
SWDIO
3.3V
J6
Conn_01x05
Break Out Communication
CAN_H
CAN_L
MOSI
MISO
CS_PI
GND
SCLK
J1
Conn_01x06
J7
Mega Connector/Breakout (if we have space)
nRST
GATE1
GATE2
GATE3
GATE4
MOSI
SCLK
MISO
GATE5
GATE6
GATE7
GATE8
GATE9
GATE10
STM_TO_INPUT_CAN
STM_TO_OUTPUT_CAN
PWM_A1
PWM_A2
PWM_B1
PWM_B2
PWM_C1
PWM_C2
PWM_D1
PWM_D2
SWCLK
SWDIO
SDA
SCL
CS_BMI
nINT_BN0
RST_BN0
INT1_BMI
Sheet: /Connector_Signals/
File: connector_signals.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5-unknown
Id: 7/6